



MODULE NAME:	MODULE CODE:
SYSTEM ANALYSIS AND DESIGN	SAND6211
SYSTEM ANALYSIS AND DESIGN	SAND6211d
SYSTEM ANALYSIS AND DESIGN	SAND6211p

ASSESSMENT TYPE:	TEST (PAPER ONLY)
TOTAL MARK ALLOCATION:	60 MARKS
TOTAL HOURS:	1 HOUR (+5 minutes reading time)
INSTRUCTIONS: 1. Please adhere to all instructions in the assessment booklet. 2. Independent work is required. 3. Five minutes per hour of the assessment to a maximum of 15 minutes is dedicated to reading time before the start of the assessment. You may make notes on your question paper, but not in your answer sheet. Calculators may not be used during reading time. 4. You may not leave the assessment venue during reading time, or during the first hour or during the last 15 minutes of the assessment. 5. Ensure that your name is on all pieces of paper or books that you will be submitting. Submit all the pages of this assessment's question paper as well as your answer script. 6. Answer all the questions on the answer sheets or in answer booklets provided. The phrase 'END OF PAPER' will appear after the final set question of this assessment. 7. Remember to work at a steady pace so that you are able to complete the assessment within the allocated time. Use the mark allocation as a guideline as to how much time to spend on each section.	
Additional instructions: 1. This is a CLOSED BOOK assessment. 2. Calculators are not allowed. 3. For multiple-choice questions, give only one (1) response per question. The marker will ignore any question with more than one answer, unless otherwise stated. You should, therefore, be sure of your answer before committing it to paper. 4. Answer All Questions.	

Question 1**(Marks: 10)**

Multiple-choice questions: Select one correct answer for each of the following unless stated otherwise. In your answer booklet, write down only the number of the question and next to it, the letter of the correct answer (s).

Q.1.1	Which one of the diagrams below can be used to document the details of a Use Case diagram?	(1)
	(a) Class diagram;	
	(b) Data flow diagram;	
	(c) Workflow diagram;	
	(d) Entity relationship diagram;	
	(e) Context diagram.	
Q.1.2	_____ can be describing as a planned undertaking that has a beginning and an end and produces some definite result?	(1)
	(a) Use case;	
	(b) Activity;	
	(c) Project;	
	(d) Node;	
	(e) Object class.	
Q.1.3	_____ is a set of interrelated computer components that collects, stores, and provides as output the information needed to complete business tasks.	(1)
	(a) Information system;	
	(b) Computer Application;	
	(c) Agile development;	
	(d) System Analysis;	
	(e) None of the above.	

Q.1.4	Which one of the following activities can be considered a Pre-project activity?	(1)
	(a) Determining the major components needed for the project;	
	(b) Obtaining budget approval;	
	(c) Defining iterations and assigning functional areas to iterations;	
	(d) Determining team members and responsibilities;	
	(e) Modelling system requirements.	
Q.1.5	Which of the options below do not represent a section that can be found in a system vision document?	(2)
	(a) Problem description;	
	(b) System capabilities;	
	(c) Business benefits;	
	(d) Use cases;	
	(e) Domain classes.	
Q.1.6	Day 1 of a project is, in essence, a planning day. Which three activities form part of the planning day activities?	(3)
	(a) Determine the major components;	
	(b) Design the database structure;	
	(c) Define the iterations;	
	(d) Design the system's high-level structure;	
	(e) Determine the team members and their responsibilities.	
Q.1.7	_____ is a computer software program that executes on a computing device to carry out a specific function or set of related functions.	(1)
	(a) Agile Development;	
	(b) Iterative Development;	
	(c) Projects;	
	(d) Computer Application (app);	
	(e) None of the Above.	

Question 2**(Marks: 35)**

The questions below relate to the following scenario:

South African Travel Center is a website that allows customers to search for and purchase flight tickets online. In order to view the flights available on a given day, a customer will need to follow the steps below:

1. The customer would specify that they would like to purchase a one way ticket.
2. The customer would then specify an airport from which they wish to depart as well as a destination.
3. A departure date would then be specified along with the number of passengers.
4. A schedule of possible flights will be displayed to the customer from which he/ she will select the one he/ she prefers.
5. The customer will then enter his/ her details along with their credit card details.
6. The payment will be processed and a printable ticket will be generated which the customer can print from the system.

Q.2.1	Draw a Use Case Diagram to represent the scenario provided at the beginning of the question.	(24)
Q.2.2	Which type of requirements (functional- or non-functional) is documented in the Use Case Diagram for the ticket purchasing system? Motivate your answer.	(2)
Q.2.3	If the website were to make provision for payments to also be made using internet banking or EFT, how could the relationship between payment, credit card payment and EFT payment be represented using domain class modelling notation?	(5)
Q.2.4	List the four steps one would follow in order to develop a System Sequence.	(4)

Question 3		(Marks: 15)
For Question 3 , refer back to the scenario presented at the beginning of Question 2 .		
Q.3.1	Identify each components of the three-layer architecture used by the ticket purchasing system presented at the beginning of Question2 .	(6)
Q.3.2	There are several critical issues that must be considered in the deployment of any new system for external use. List any <u>three</u> of these issues.	(3)
Q.3.3	What is the importance of feedback in the design of interfaces?	(2)
Q.3.4	Suggest <u>two</u> ways in which feedback can be used in the ticket purchasing system.	(2)
Q.3.5	Designing Web and app-based user interfaces for handheld devices presents several Challenges. List any <u>two</u> such challenges.	(2)

END OF PAPER